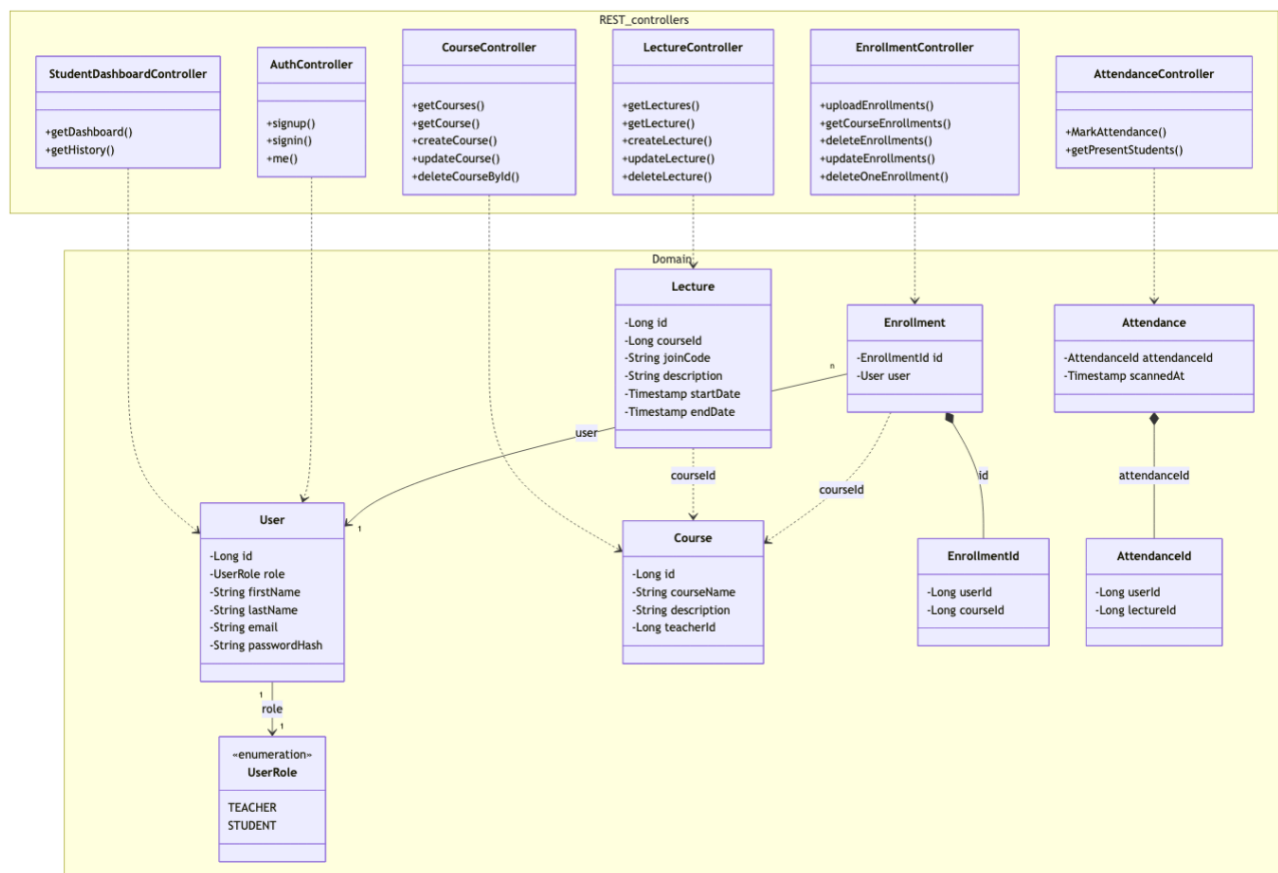


Exercise: Class Diagram

Class diagram: The diagram shows domain entities and application (REST) controller classes. Instance variables and key public operations are included where they clarify behaviour. User-interface classes are not part of this backend and are excluded. Details such as repositories, services, DTO field lists, configuration, and Lombok-generated builders are omitted to keep the view readable, as required.



Class diagram: domain model and REST controllers.

Supporting description:

Purpose: The software supports attendance management for teaching: teachers maintain courses and lectures, students are linked to courses through enrollments, and attendance records who was present at a given lecture and when the check-in occurred.

Domain classes: User represents people using the system (e.g. teachers and students). The UserRole enumeration distinguishes TEACHER and STUDENT. Password material is stored as a hash, not plain text.

Course describes a module: name, description, and teacherId, which identifies the owning teacher.

Lecture is one scheduled session of a course. It references a course via courseId, exposes a joinCode for student check-in, and has a time window (startDate, endDate).

Enrollment models the fact that a student is registered on a course. Its identity is a composite EnrollmentId (userId, courseId). The entity also references User (the enrolled student).

Attendance records that a user attended a lecture. The composite AttendanceId pairs userId and lectureId; scannedAt stores when the attendance was recorded.

Associations on the diagram reflect these relationships: role of a user, composition of entities with their composite keys, enrollment of a user in a course (via ids), lectures belonging to a course (via courseId), and attendance keyed by user and lecture.

Application controller classes:

The REST controllers are the application's HTTP entry points:

Controller	Responsibility (summary)
AuthController	Registration (signup), authentication (signin), and returning the current user profile (me).
CourseController	Listing and CRUD for courses for the authenticated teacher context.
EnrollmentController	Bulk upload of enrollments (e.g. XML), listing and deleting enrollments for a course, updating a single student's enrollment.
LectureController	Listing, reading, creating, updating, and deleting lectures (optionally filtered by course).
AttendanceController	Recording attendance using a lecture join code and listing students present for a lecture.
StudentDashboardController	Student-facing dashboard and history of attendance.

Dashed dependencies from controllers to domain types indicate which main concepts each controller operates on. Actual HTTP paths, request/response DTOs, and security (e.g. JWT) are implemented in code but are not expanded here to avoid excessive detail.

Scope and exclusions: The diagram intentionally does not show: separate UI layers (there is no UI in this repository), Spring Data repositories, service classes that sit between controllers and persistence, DTO classes (only referenced by name in operation signatures where helpful), security configuration, or test classes. This matches the instruction to keep the class diagram clear and illustrative.